Jonathan Sarasohn

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PROFESSIONAL EXPERIENCE

Stryker, *Project Engineer, Advanced Operations Global Engineering Development* | Mahwah, NJ *September 2024 - Present*

- Led a cross-functional root-cause investigation into a critical manufacturing issue, coordinating technical analysis across manufacturing, quality, biocompatibility, and regulatory to implement corrective process changes and reduce scrap by over 30%
- Designed, prototyped, and evaluated orthopedic implants and surgical instruments through multiple rounds of laboratory and cadaveric testing, using performance and usability findings to drive design iterations
- Supported product development and commercialization through Voice of Customer/Voice of Sales (VOC) analysis, survey development, pre-commercial planning, and operative-technique creation
- Lead supplier and manufacturing process development for additively manufactured and injection-molded components, including design for manufacturability (DFM), design changes, prototyping, cleaning process development, and design transfer
- Developed digital engineering tools utilizing the Microsoft Power platform and AI tools, including a purchase-order request application with KPI tracking, a VOC analyzer, and an automated drawing management tool
- Completed rotations across Advanced Operations, R&D, and Upstream Marketing, contributing to technical peer reviews, DFMECA, CQA assessments, product claims documentation, and design transfer

Oishii, *Robotics Engineering Intern* | Jersey City, NJ *May 2022 - August 2024*

- Designed, integrated, and tested the mechanical, electrical, and embedded systems for a custom robot and robotic beehives
- Reduced robotic system assembly time by 98% through a redesign using Design For Assembly (DFA) principles
- Developed a new electrical architecture incorporating custom PCBs and a DC power distribution system
- Developed and deployed embedded firmware using Arduino, Raspberry Pi, and Balena for remotely deployed robotic systems
- Managed component sourcing and 2 contractors through the assembly of 100+ robotic beehives for a new vertical farm

Stryker, *Continuous Improvement Intern* | Mahwah, NJ *May 2023 - August 2023*

- Conceived and implemented a custom digital throughput and cycle time tracking solution to enhance production line visibility
- Realized annual savings of over $300,000 through process enhancements within a single manufacturing cell
- Spearheaded two KATA projects to identify and remedy critical capacity bottlenecks, resulting in a 30% boost in throughput

LEADERSHIP EXPERIENCE

Binghamton University Mars Rover, *Chief Mechanical Engineer* | Binghamton, NY *April 2023 - June 2024*

- Led an 80+ member multidisciplinary team in designing and building a Mars Rover for the international University Rover Challenge
- Directed six sub-team leads and coordinated rover-level design across the mechanical, electrical, and controls teams
- Oversaw the team's business operations including finance, sponsorships, and marketing, raising over $60k in sponsorships
- Placed 20th internationally out of over 150 universities competing

Mechanical Team Member/Lead | Binghamton, NY *September 2020 - June 2023*

- Led weekly design reviews of Fusion 360 CAD models and developed mechanical solutions for the robotic arm
- Established team-wide mechanical policies, best practices, and conducted information sessions, while also creating and spearheading the mechanical onboarding program, successfully training 40+ new mechanical engineers
- Designed custom 400:1 reduction cycloidal gearboxes and participated in design reviews with team leadership and technicians

PATENTS

- **Pollination System,** *US 12433211 B2* | Assignee: Oishii *Issued: October 2025*

EDUCATION

Binghamton University, SUNY

Bachelor of Science in Mechanical Engineering *May 2024*

Bachelor of Arts in Geography, Concentration in Urban and Regional Planning *May 2024*

Honors: Binghamton University Scholars Program, Pi Tau Sigma, Gamma Theta Upsilon

VOLUNTEER EXPERIENCE

Frisch Robotics Program, *Coach* | Paramus, NJ *October 2024 - Present*

- Mentor high school students in FIRST Robotics, providing technical guidance in robot design, engineering, and problem-solving
- Support robotics curriculum development and instruction, advising on program structure, budgeting, and administration

Binghamton University Mars Rover, *Advisor* | Binghamton, NY *June 2024 - Present*

- Advise Binghamton's Mars Rover Team on mechanical engineering, rover design, project management, and team administration

COVID-19 Personal Protective Equipment Manufacturing, *Project Lead* | Teaneck, NJ *April 2020 - August 2020*

- Designed and prototyped a low-cost, manufacturable face shield to address PPE shortages among frontline healthcare personnel
- Led material sourcing, team coordination, budgeting, and stakeholder communication to deliver 700+ face shields

TECHNICAL SKILLS

Medical Device: ISO 13485, Design Controls, Design Transfer, DFMECA, Root Cause Analysis, DFM/DFA, Tolerance Analysis
CAD & Analysis: Creo, SolidWorks, Autodesk Inventor, Fusion 360, GD&T, Finite Element Analysis, Windchill
Software & Automation: MATLAB, Python, Power BI, Power Automate, Power Apps
Manufacturing: Additive Manufacturing, CAM, CNC Machining, Milling, Waterjet, Welding, Optical Comparator, Guage Pins